

Project 2: Linux System Security Audit

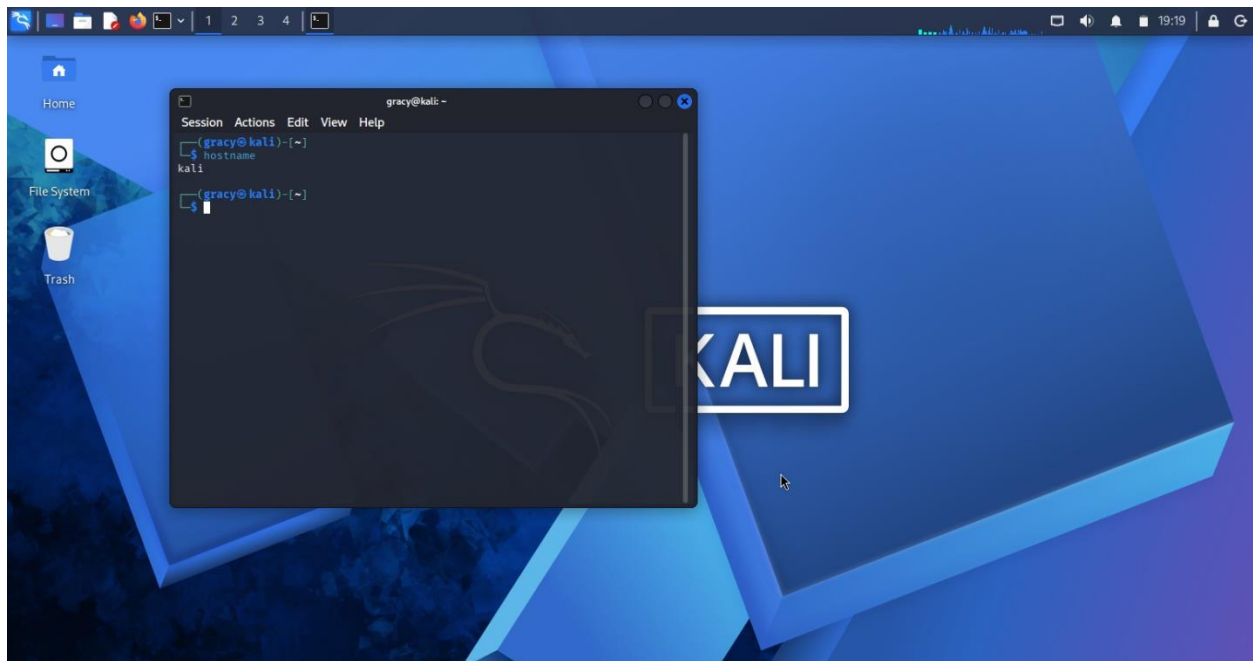
Objective: Perform a basic security audit on a Linux system using Linux commands to understand the file system, permissions, users, processes, logs and troubleshooting.

Tools used:

- Kali Linux

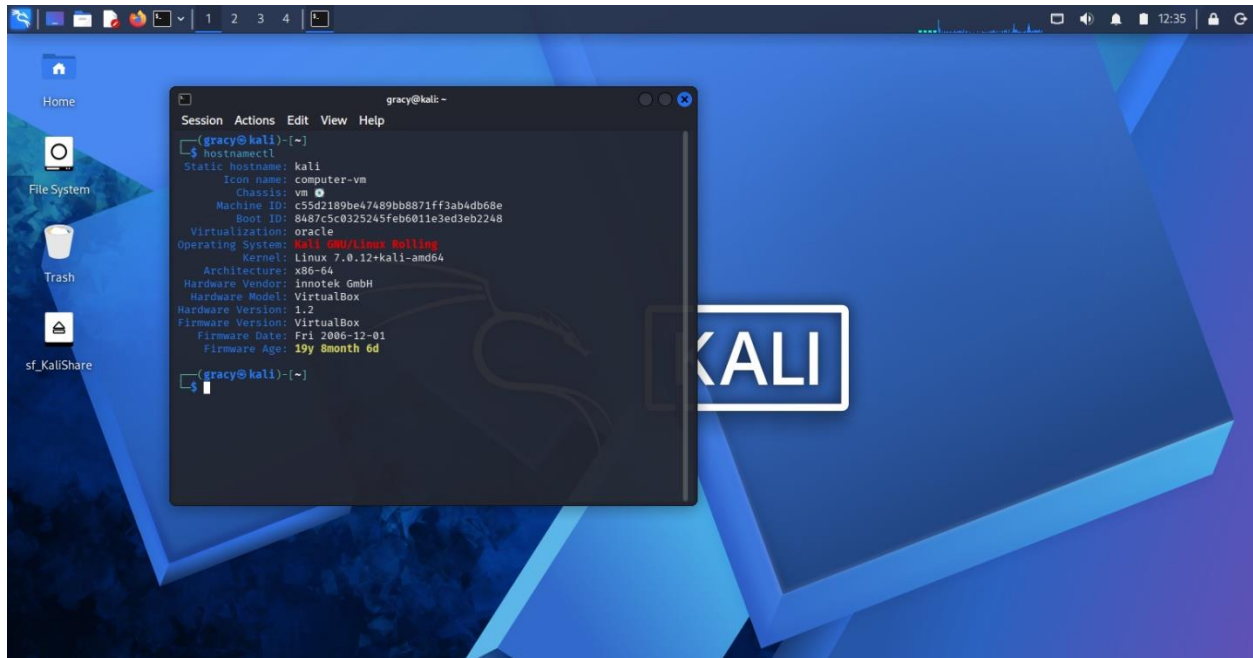
Task 1: System Information

Command 1: hostname



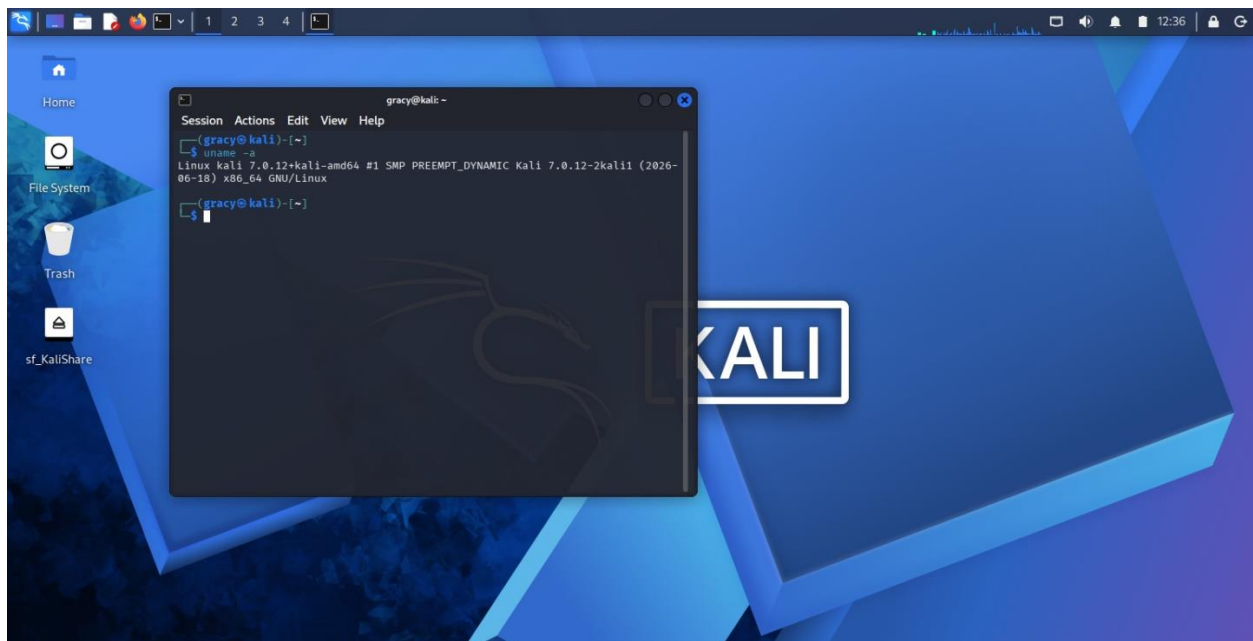
Explanation: Displays the hostname (computer name) of the Linux system.

Command 2: hostnamectl



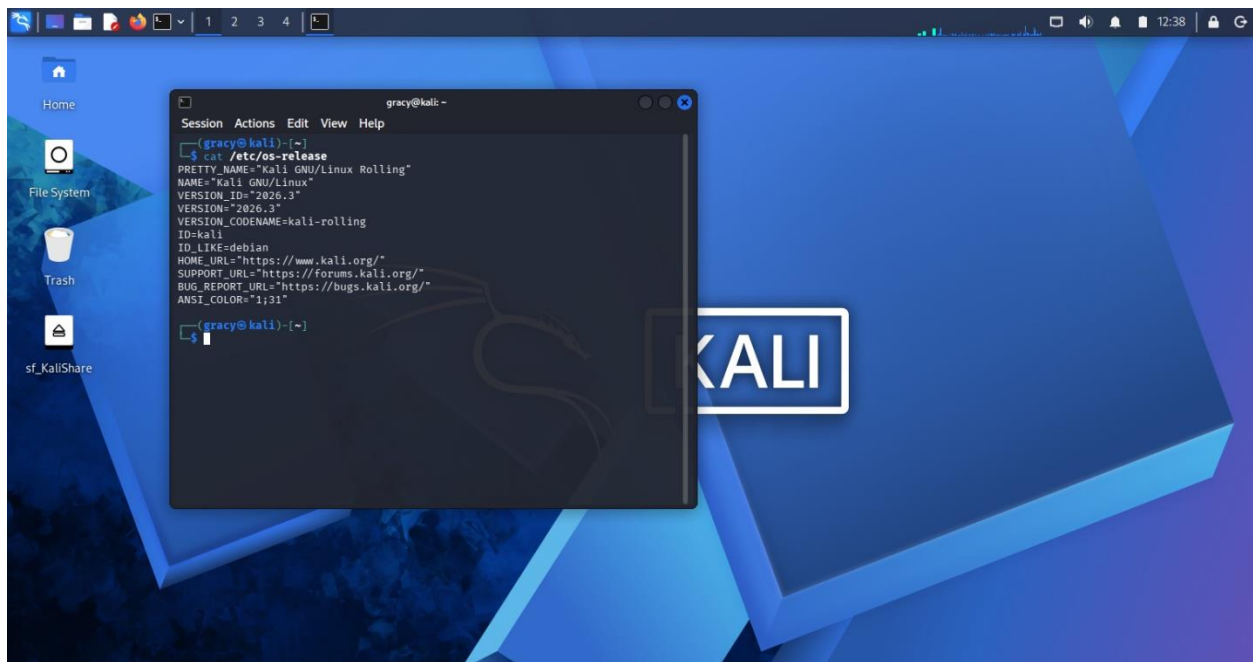
Explanation: Displays detailed system information, including the hostname, operating system, kernel version, architecture, and hardware details.

Command 3: uname -a



Explanation: Displays complete system information, including the kernel name, kernel version, machine architecture, and operating system.

Command 4: `cat /etc/os-release`

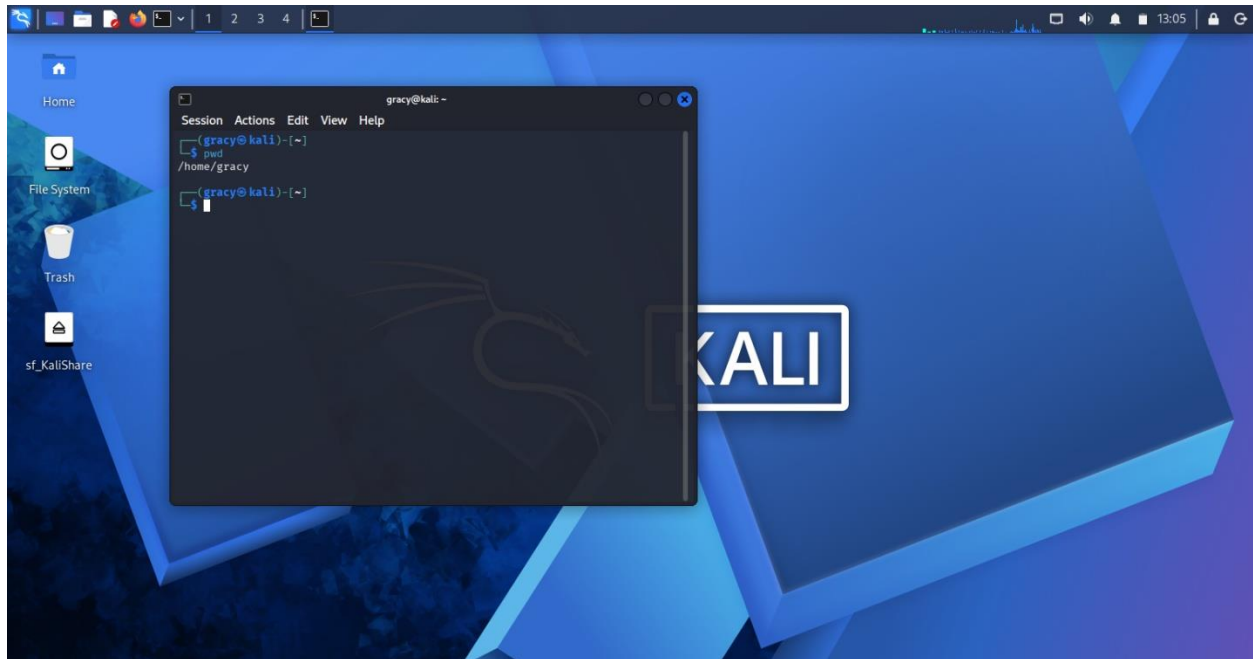
A screenshot of a Kali Linux desktop environment. The desktop has a blue and purple geometric background with a large 'KALI' logo. On the left side, there are icons for 'Home', 'File System', 'Trash', and 'sf_KaliShare'. In the center, a terminal window titled 'gracy@kali: ~' is open. The terminal shows the command 'cat /etc/os-release' and its output, which includes system information like 'PRETTY_NAME=Kali GNU/Linux Rolling', 'NAME=Kali GNU/Linux', 'VERSION_ID=2026.3', 'VERSION_CODENAME=kali-rolling', 'ID=kali', 'ID_LIKE=debian', 'HOME_URL=https://www.kali.org/', 'SUPPORT_URL=https://forums.kali.org/', 'BUG_REPORT_URL=https://bugs.kali.org/', and 'ANSI_COLOR=1;31'. The terminal prompt is '\$'.

Explanation: Displays Linux distribution information such as the distribution name, version, ID, and other operating system details.

Observation: The commands successfully displayed the system hostname, operating system details, Linux kernel version, architecture, and distribution information. This information helps identify the system configuration before performing a security audit.

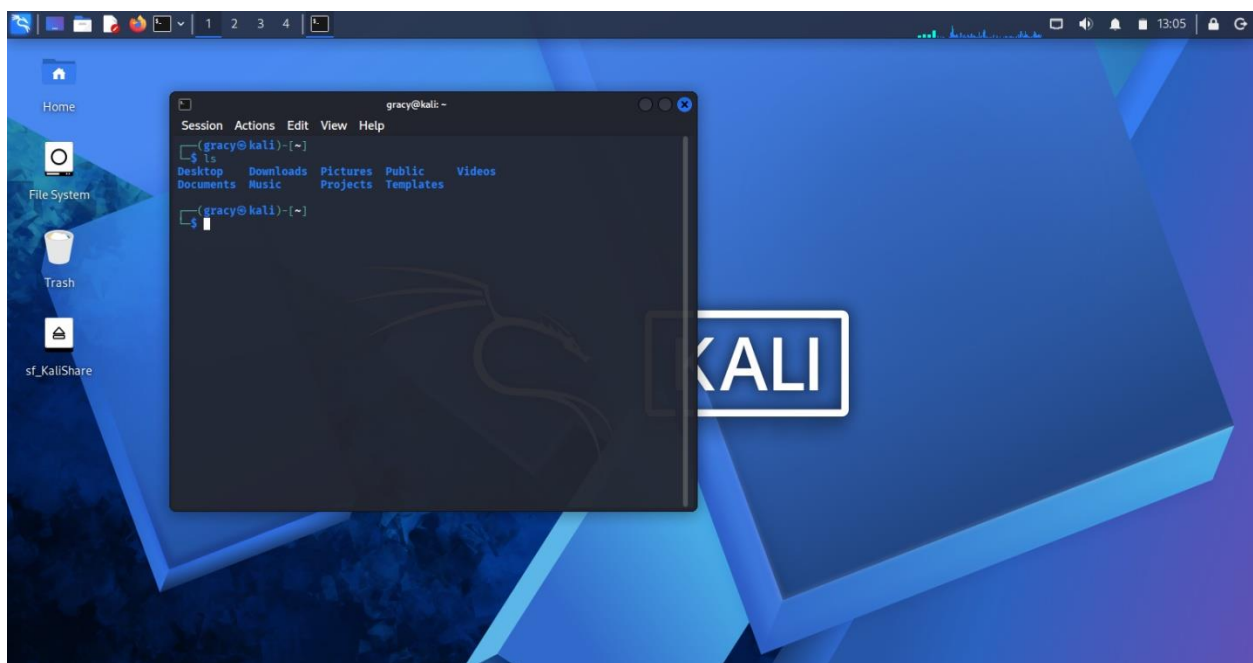
Task 2: File System Navigation

Command 1: pwd



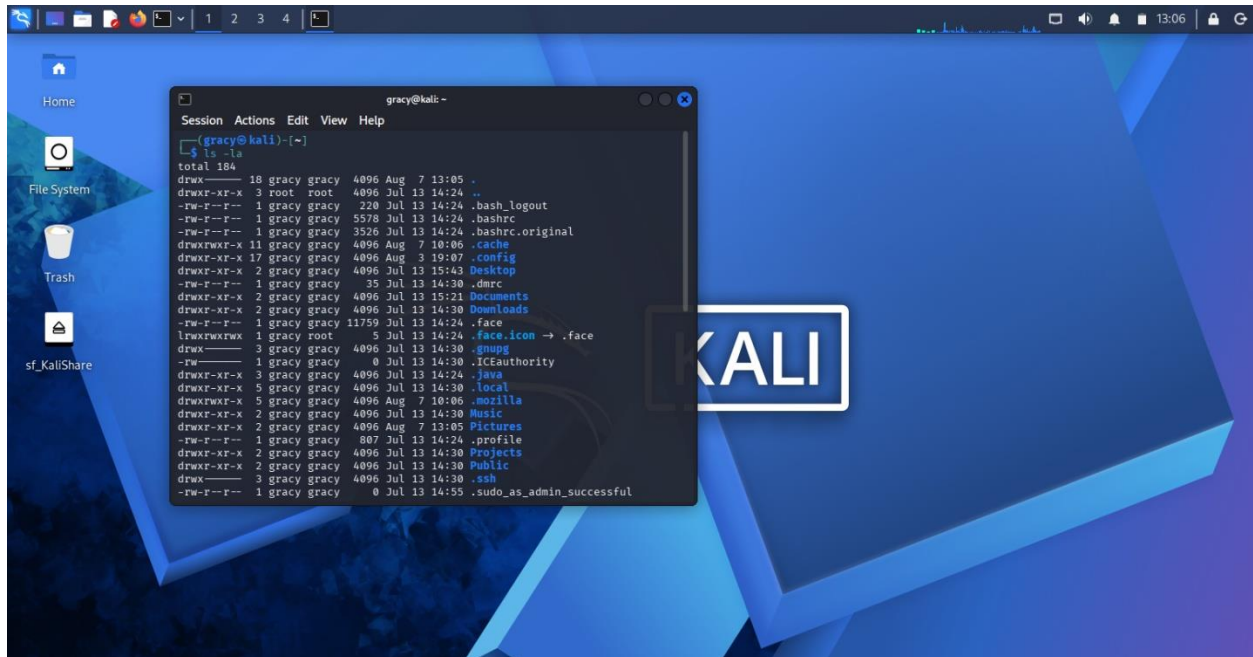
Explanation: Displays the current working directory.

Command 2: ls



Explanation: Lists the files and directories in the current location.

Command 3: `ls -la`

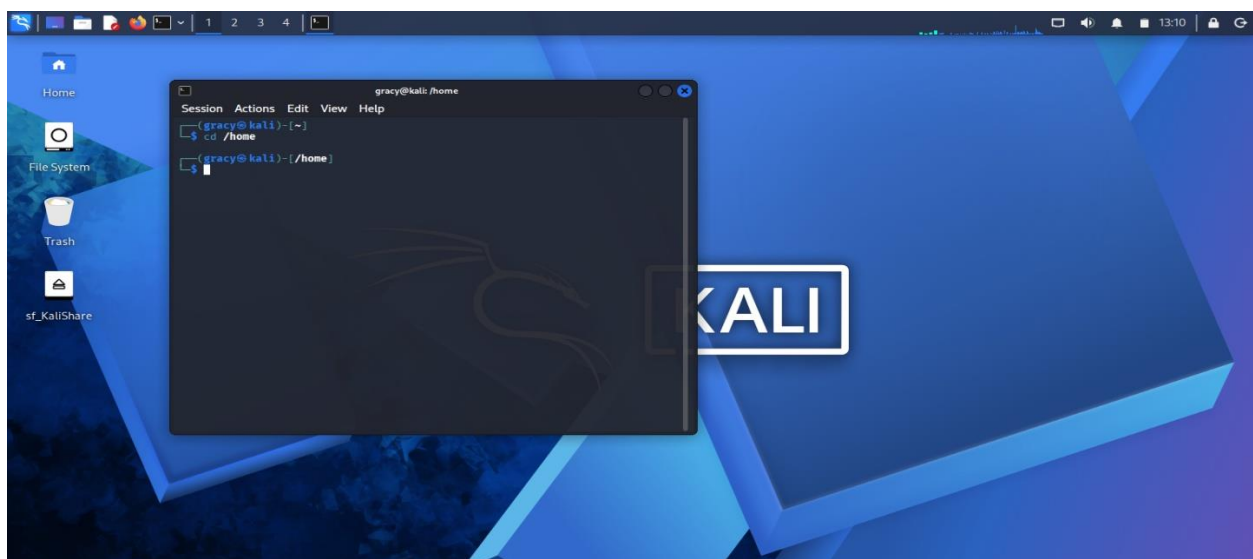


The screenshot shows a Kali Linux desktop with a blue background and a large 'KALI' logo. A terminal window is open, displaying the output of the `ls -la` command. The output lists files and directories with their permissions, owner, size, and modification date. The files listed include `..`, `.bash_logout`, `.bashrc`, `.bashrc.original`, `.cache`, `.config`, `Desktop`, `.dmrc`, `Documents`, `Downloads`, `.face`, `.face.icon`, `.gnupg`, `.ICEauthority`, `.java`, `.local`, `.mozilla`, `Music`, `Pictures`, `.profile`, `Projects`, `Public`, `.ssh`, and `.sudo_as_admin_successful`.

```
gracy@kali: ~  
$ ls -la  
total 184  
drwxr-xr-x 18 gracy gracy 4096 Aug  7 13:05 .  
drwxr-xr-x  3 root  root  4096 Jul 13 14:24 ..  
-rw-r--r--  1 gracy gracy  220 Jul 13 14:24 .bash_logout  
-rw-r--r--  1 gracy gracy 5578 Jul 13 14:24 .bashrc  
-rw-r--r--  1 gracy gracy 3526 Jul 13 14:24 .bashrc.original  
drwxrwxr-x 11 gracy gracy 4096 Aug  7 10:06 .cache  
drwxr-xr-x 17 gracy gracy 4096 Aug  3 19:07 .config  
drwxr-xr-x  2 gracy gracy 4096 Jul 13 15:43 Desktop  
-rw-r--r--  1 gracy gracy   35 Jul 13 14:30 .dmrc  
drwxr-xr-x  2 gracy gracy 4096 Jul 13 15:21 Documents  
drwxr-xr-x  2 gracy gracy 4096 Jul 13 14:30 Downloads  
-rw-r--r--  1 gracy gracy 11759 Jul 13 14:24 .face  
lrwxrwxrwx  1 gracy root    5 Jul 13 14:24 .face.icon -> .face  
drwxr-xr-x  3 gracy gracy 4096 Jul 13 14:30 .gnupg  
-rw-r--r--  1 gracy gracy   60 Jul 13 14:30 .ICEauthority  
drwxr-xr-x  3 gracy gracy 4096 Jul 13 14:24 .java  
drwxr-xr-x  5 gracy gracy 4096 Jul 13 14:30 .local  
drwxrwxr-x  5 gracy gracy 4096 Aug  7 10:06 .mozilla  
drwxr-xr-x  2 gracy gracy 4096 Jul 13 14:30 Music  
drwxr-xr-x  2 gracy gracy 4096 Aug  7 13:05 Pictures  
-rw-r--r--  1 gracy gracy  807 Jul 13 14:24 .profile  
drwxr-xr-x  2 gracy gracy 4096 Jul 13 14:30 Projects  
drwxr-xr-x  2 gracy gracy 4096 Jul 13 14:30 Public  
drwxr-xr-x  3 gracy gracy 4096 Jul 13 14:30 .ssh  
-rw-r--r--  1 gracy gracy   60 Jul 13 14:55 .sudo_as_admin_successful
```

Explanation: Lists all files and directories, including hidden files, along with detailed information such as permissions, owner, size, and modification date.

Command 4: `cd /home`

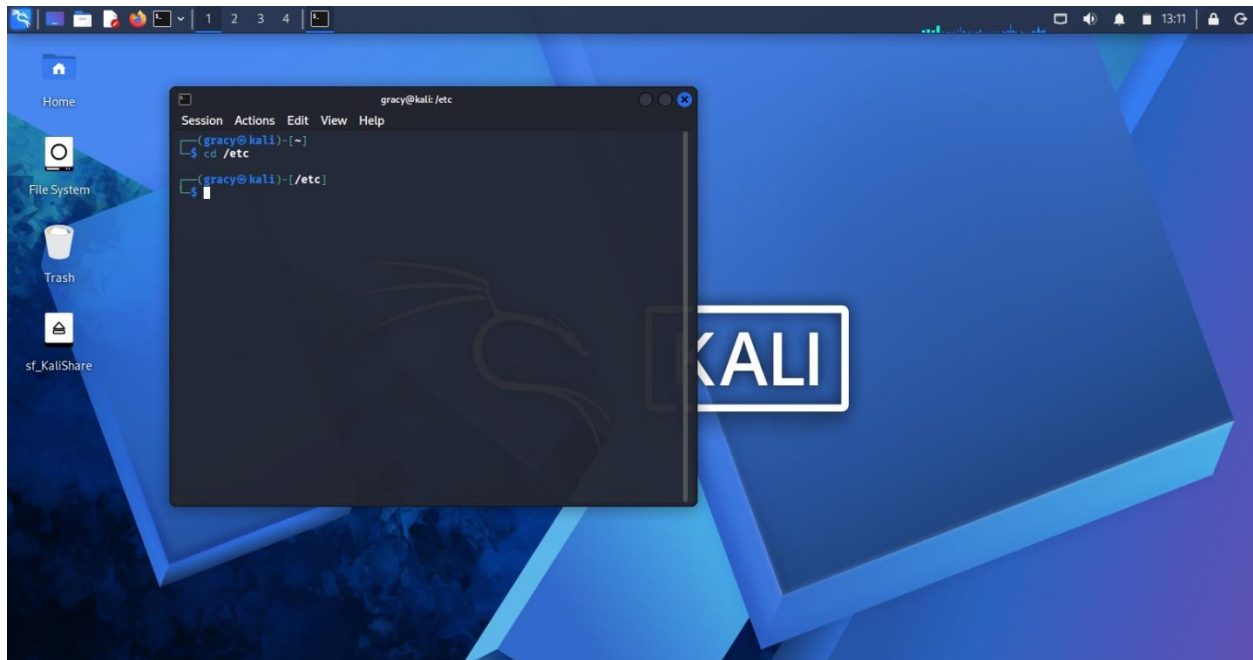


The screenshot shows the same Kali Linux desktop environment. A terminal window is open, displaying the output of the `cd /home` command. The prompt has changed from `gracy@kali: ~` to `gracy@kali: /home`, indicating the current directory has been changed to `/home`.

```
gracy@kali: /home  
$ cd /home  
gracy@kali: /home
```

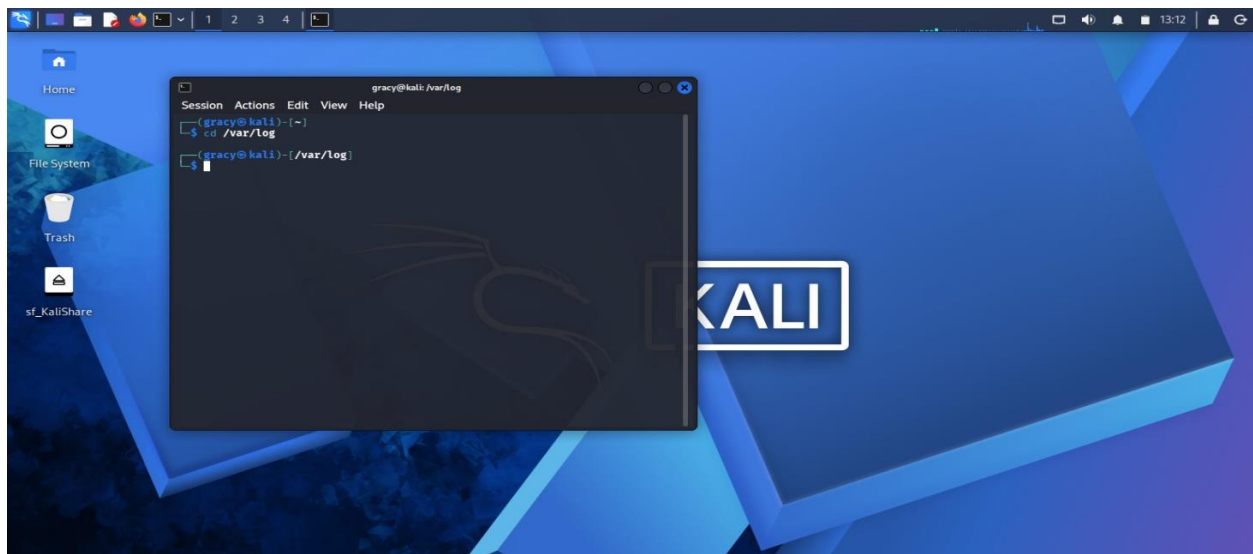
Explanation: Changes the current directory to /home, which contains the home directories of users.

Command 5: `cd /etc`



Explanation: Changes the current directory to /etc, which stores system configuration files.

Command 6: `cd /var/log`

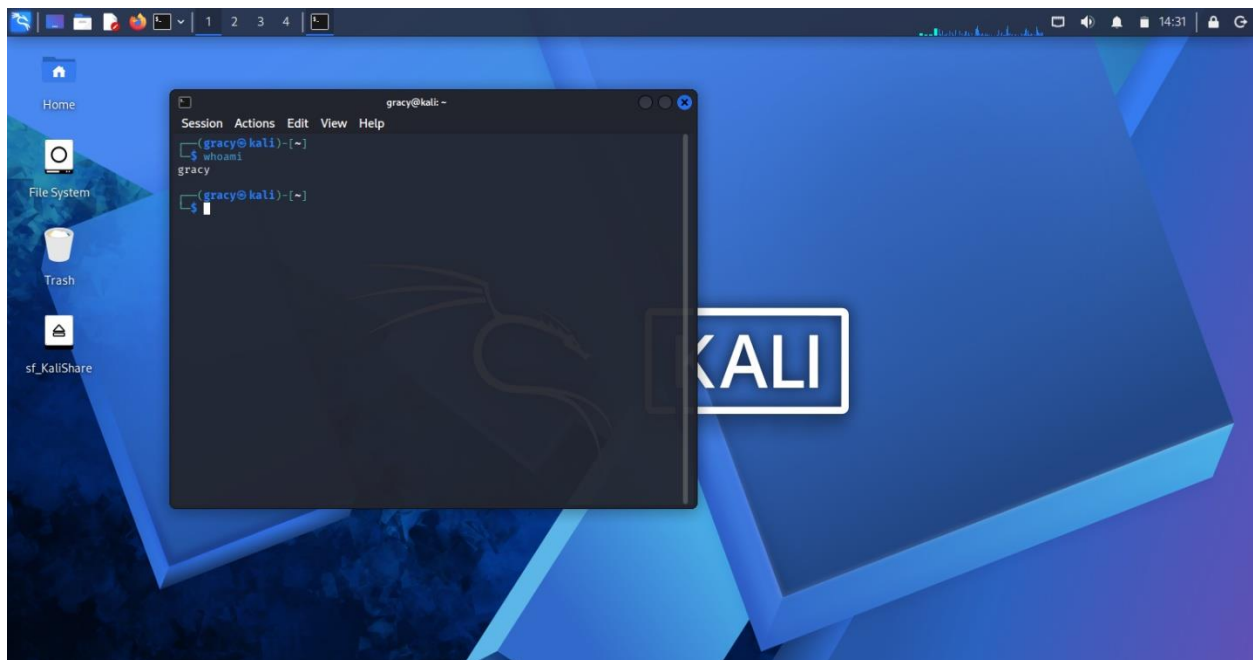


Explanation: Changes the current directory to /var/log, where system log files are stored.

Observation: The commands were successfully executed. They helped navigate different directories, view files (including hidden files), and understand the purpose of important Linux directories such as /home, /etc, and /var/log.

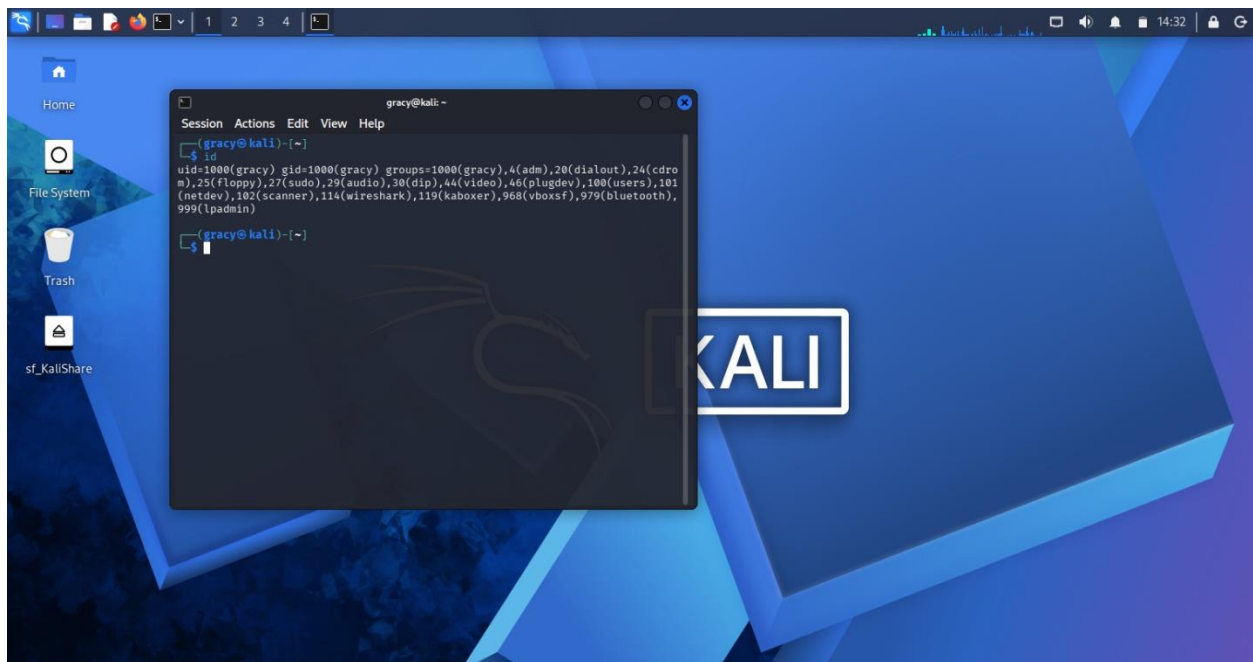
Task 3: Users and Groups

Command 1: whoami



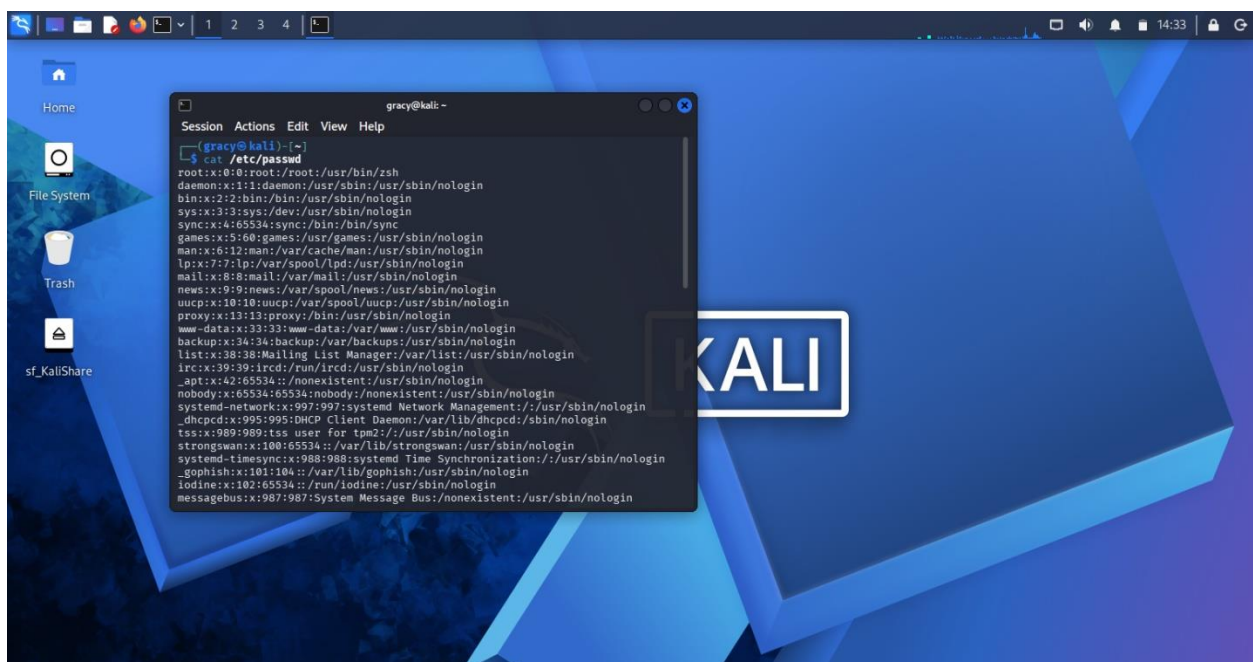
Explanation: Displays the username of the currently logged-in user.

Command 2: id



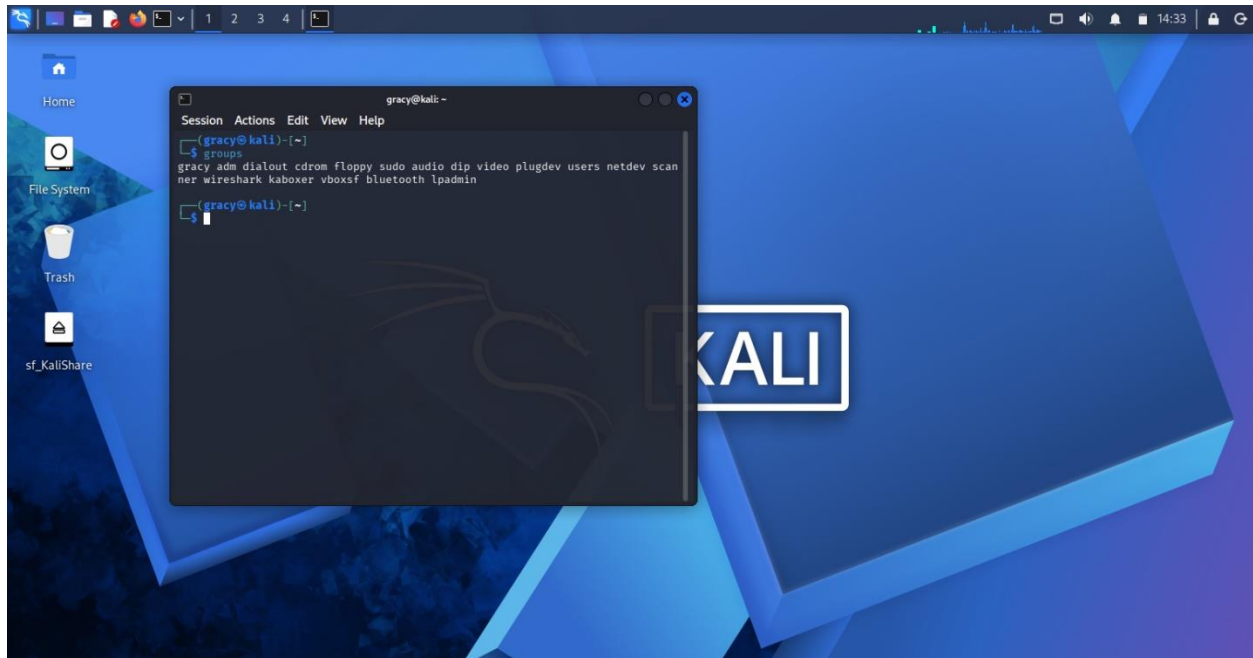
Explanation: Displays the current user's UID, GID, and group memberships.

Command 3: cat /etc/passwd



Explanation: Displays all user accounts stored on the Linux system.

Command 4: groups



Explanation: Displays the groups to which the current user belongs.

Identification:

Current User: gracy

User ID: 1000

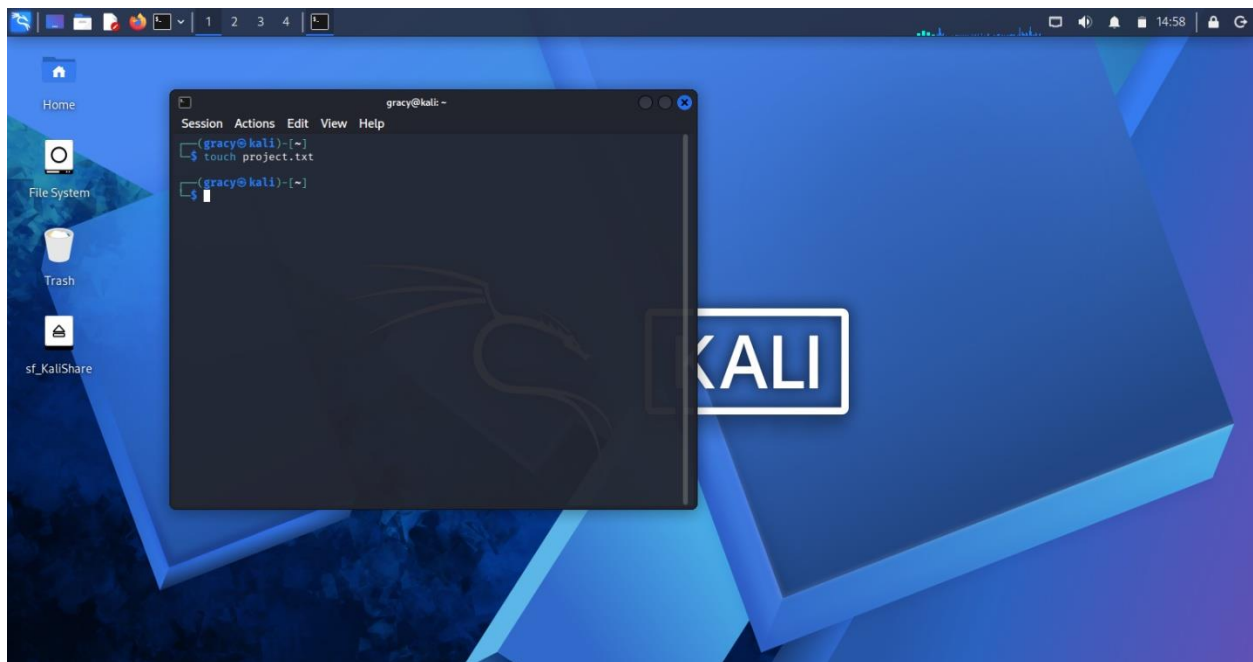
Group ID: 1000

Available Users: root, daemon, bin, sys, ... etc.

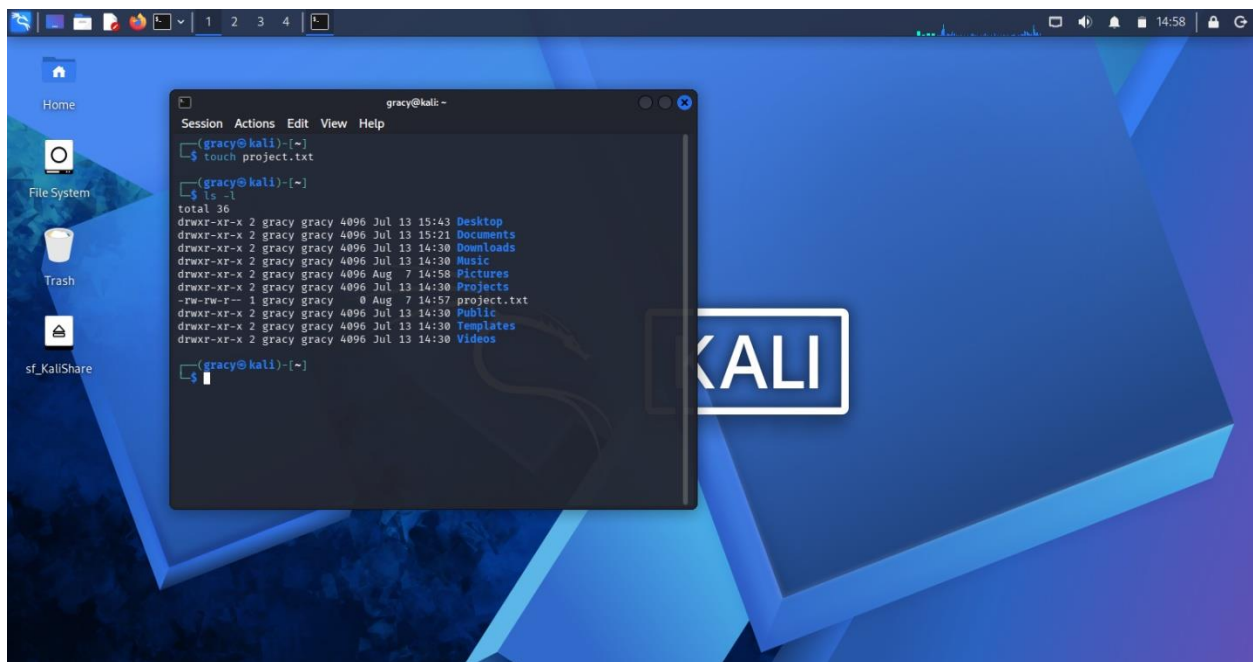
Task 4: File Permissions

Command 1: touch project.txt

Explanation: Creates a new empty file named project.txt.

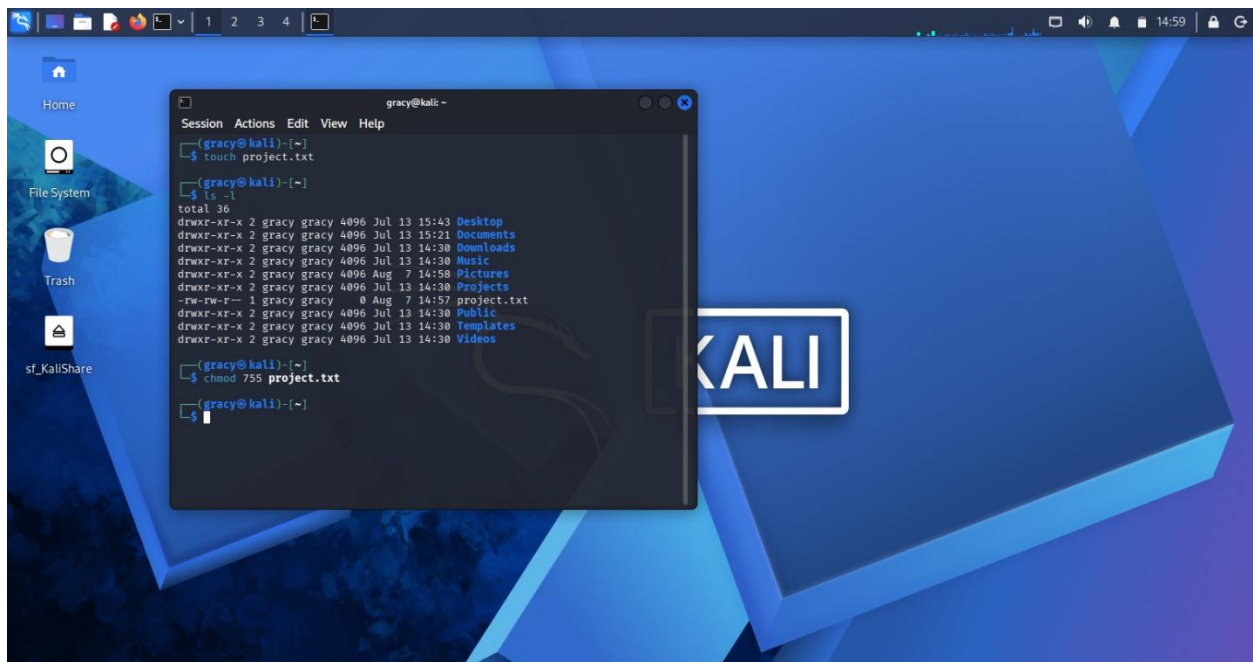


Command 2: ls -l



Explanation: Displays the file permissions, owner, group, file size, and other details.

Command 3: chmod 755 project.txt



Explanation: Changes the file permissions to 755:

- Owner: Read, Write, Execute (rwx)
- Group: Read, Execute (r-x)
- Others: Read, Execute (r-x)

Command 4: `ls -l`

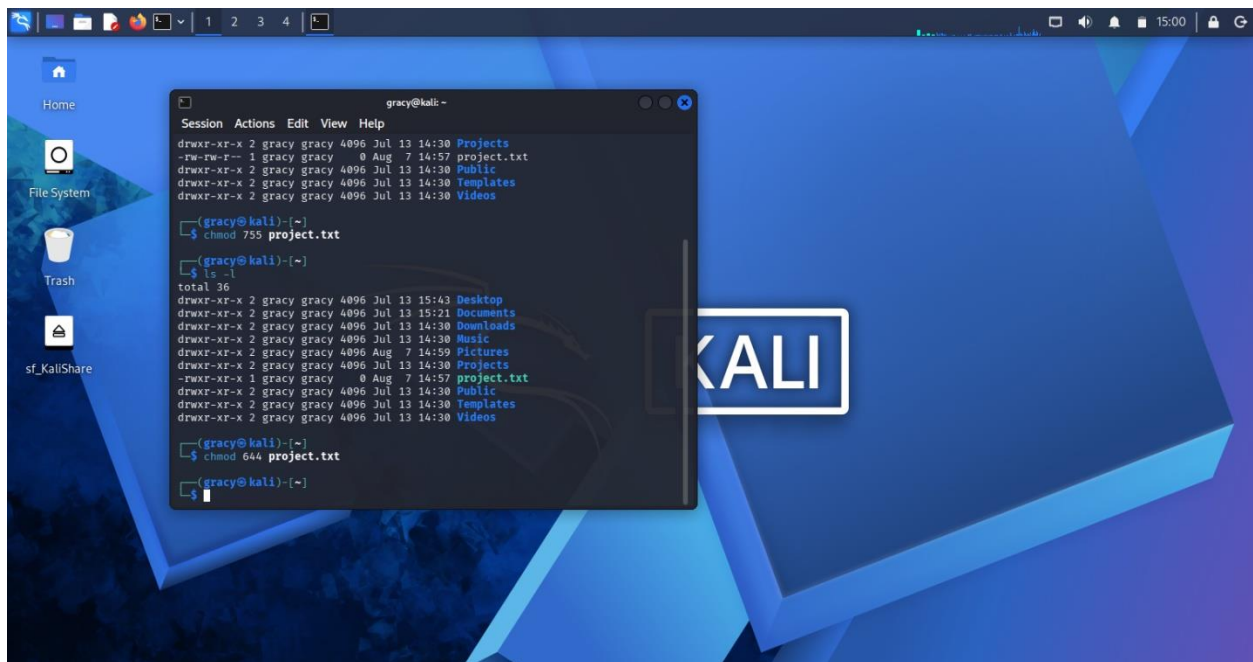
(Same as Command 2)

Command 5: `chmod 644 project.txt`

Explanation:

Changes the file permissions to 644:

- Owner: Read, Write (rw-)
- Group: Read (r--)
- Others: Read (r--)



Command 6: ls -l

(Same as Command 2)

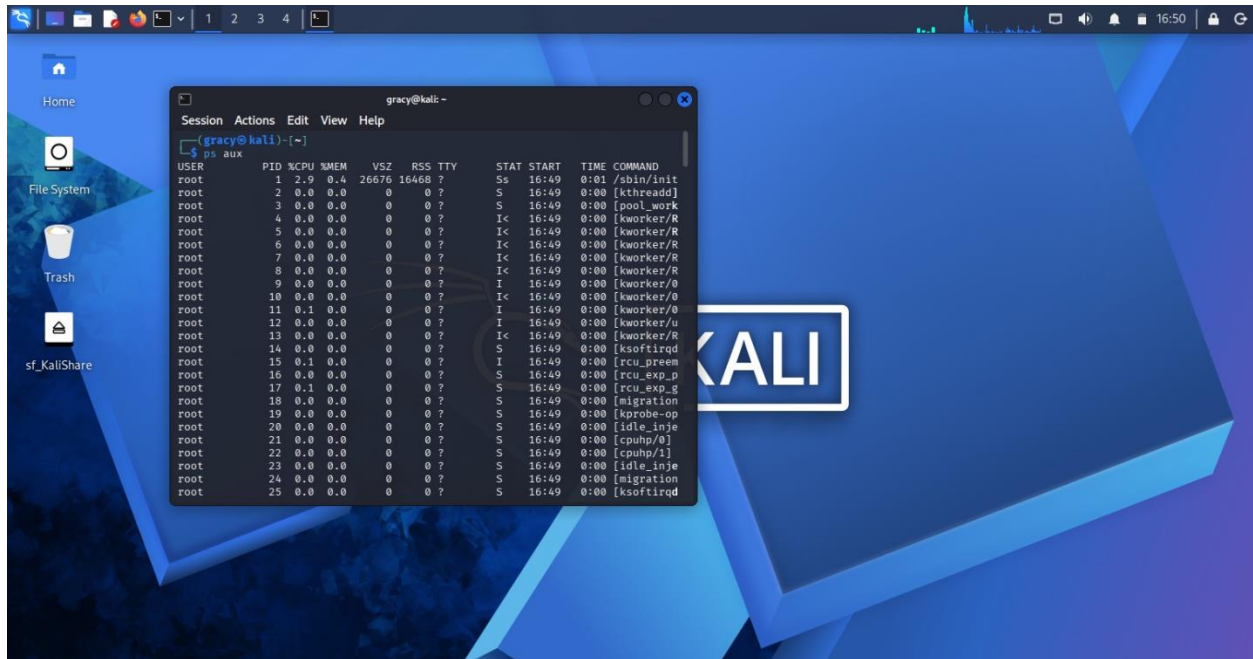
Read, Write and Execute Permissions

- **Read (r):** Allows viewing the contents of a file.
- **Write (w):** Allows modifying or editing the file.
- **Execute (x):** Allows executing the file as a program or script.

Observation: The project.txt file was created successfully. The ls -l command displayed the file permissions before and after using chmod. The permissions changed correctly from the default settings to **755** and then to **644**.

Task 5: Process Monitoring

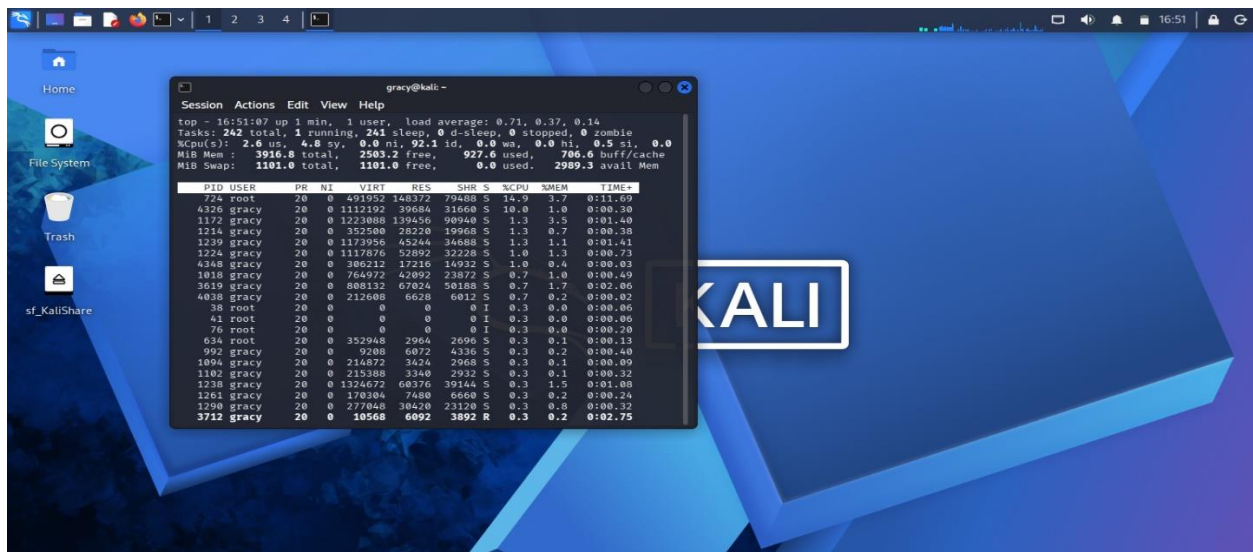
Command 1: ps aux



```
gracy@kali: ~  
$ ps aux  
USER          PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND  
root           1  2.9  0.4 26676 16468 ?        Ss   16:49   0:01 /sbin/init  
root           2  0.0  0.0      0     0 ?        S    16:49   0:00 [kthreadd]  
root           3  0.0  0.0      0     0 ?        S    16:49   0:00 [pool_work  
root           4  0.0  0.0      0     0 ?        I<   16:49   0:00 [kworker/R  
root           5  0.0  0.0      0     0 ?        I<   16:49   0:00 [kworker/R  
root           6  0.0  0.0      0     0 ?        I<   16:49   0:00 [kworker/R  
root           7  0.0  0.0      0     0 ?        I<   16:49   0:00 [kworker/R  
root           8  0.0  0.0      0     0 ?        I<   16:49   0:00 [kworker/R  
root           9  0.0  0.0      0     0 ?        I    16:49   0:00 [kworker/0  
root          10  0.0  0.0      0     0 ?        I<   16:49   0:00 [kworker/0  
root          11  0.1  0.0      0     0 ?        I    16:49   0:00 [kworker/0  
root          12  0.0  0.0      0     0 ?        I    16:49   0:00 [kworker/u  
root          13  0.0  0.0      0     0 ?        I<   16:49   0:00 [kworker/R  
root          14  0.0  0.0      0     0 ?        S    16:49   0:00 [ksoftirqd  
root          15  0.1  0.0      0     0 ?        I    16:49   0:00 [rcu_prem  
root          16  0.0  0.0      0     0 ?        S    16:49   0:00 [rcu_exp_d  
root          17  0.1  0.0      0     0 ?        S    16:49   0:00 [rcu_exp_g  
root          18  0.0  0.0      0     0 ?        S    16:49   0:00 [migration  
root          19  0.0  0.0      0     0 ?        S    16:49   0:00 [kprobe-op  
root          20  0.0  0.0      0     0 ?        S    16:49   0:00 [idle_inje  
root          21  0.0  0.0      0     0 ?        S    16:49   0:00 [cpuhp/0]  
root          22  0.0  0.0      0     0 ?        S    16:49   0:00 [cpuhp/1]  
root          23  0.0  0.0      0     0 ?        S    16:49   0:00 [idle_inje  
root          24  0.0  0.0      0     0 ?        S    16:49   0:00 [migration  
root          25  0.0  0.0      0     0 ?        S    16:49   0:00 [ksoftirqd
```

Explanation: Displays a list of all running processes, including the user, process ID (PID), CPU usage, memory usage, and the command associated with each process.

Command 2: top



```
gracy@kali: ~  
$ top  
top - 16:51:07 up 1 min, 1 user, load average: 0.71, 0.37, 0.14  
Tasks: 242 total, 1 running, 241 sleep, 0 d-sleep, 0 stopped, 0 zombie  
%Cpu(s): 2.0 us, 4.8 sy, 0.0 ni, 92.1 id, 0.0 wa, 0.0 hi, 0.5 si, 0.0 st  
MiB Mem : 3916.0 total, 2583.2 free, 927.6 used, 780.6 buff/cache  
MiB Swap: 1101.0 total, 1101.0 free, 0.0 used, 2989.3 avail Mem  
  
  PID USER      PR  NI    VIRT    RES    SHR S  %CPU  %MEM    TIME+  COMMAND  
  724 root        0   0 491952 148372 79480 S   14.9   3.7   0:11.60  
 4326 gracy     20   0 1112192 39684 31660 S   10.0   1.0   0:00.30  
 1172 gracy     20   0 1223088 139456 98940 S    1.3   3.5   0:01.40  
 1214 gracy     20   0 352500 28220 19960 S    1.3   0.7   0:00.38  
 1239 gracy     20   0 1173956 45244 34688 S    1.3   1.1   0:01.41  
 1224 gracy     20   0 1117676 52892 32228 S    1.0   1.3   0:00.73  
 4348 gracy     20   0 362212 17216 14922 S    1.0   0.4   0:00.02  
 1018 gracy     20   0 764972 42092 23872 S    0.7   1.0   0:00.49  
 3619 gracy     20   0 888132 67024 58188 S    0.7   1.7   0:02.00  
 4038 gracy     20   0 212688 6620 6012 S    0.7   0.2   0:00.02  
   38 root        0   0      0      0      0 I    0.3   0.0   0:00.00  
   41 root        0   0      0      0      0 I    0.3   0.0   0:00.00  
   76 root        0   0      0      0      0 I    0.3   0.0   0:00.20  
   634 root       20   0 352948 2904 2696 S    0.3   0.1   0:00.13  
   992 gracy     20   0 9208 6072 4336 S    0.3   0.2   0:00.40  
 1094 gracy     20   0 214872 3424 2968 S    0.3   0.1   0:00.09  
 1102 gracy     20   0 215388 3340 2932 S    0.3   0.1   0:00.32  
 1238 gracy     20   0 1324672 60376 39144 S    0.3   1.5   0:01.08  
 1261 gracy     20   0 170304 7480 6060 S    0.3   0.2   0:00.24  
 1250 gracy     20   0 277048 39420 23120 S    0.3   0.8   0:00.32  
 3712 gracy     20   0 10568 6092 3892 R    0.3   0.2   0:02.75
```

Explanation: Displays real-time information about running processes, including CPU usage, memory usage, process IDs, and system performance.

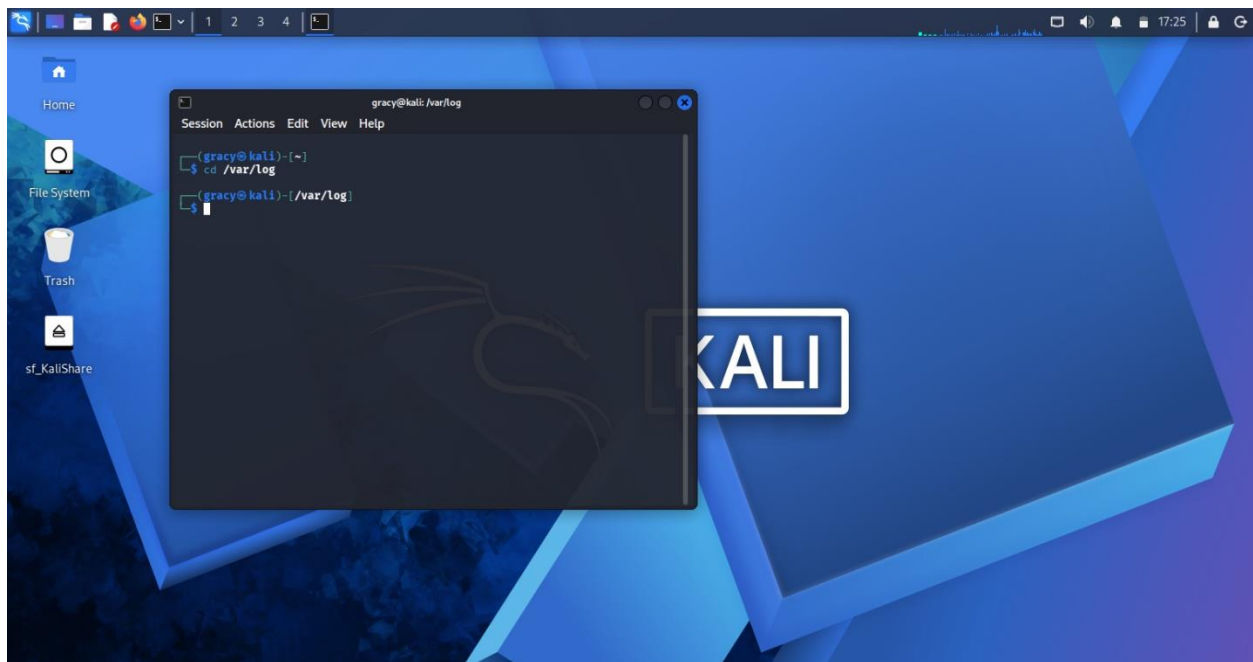
PID	User	CPU Usage	Memory Usage
724	root	14.9%	3.7%
4326	gracy	10.0%	1.0%
1172	gracy	1.3%	3.5%

Observation: The ps aux and top commands successfully displayed all running processes along with their CPU and memory usage. The top command provided real-time system performance information, making it easy to identify the processes consuming the highest CPU and memory resources. This helps in monitoring system performance and detecting resource-intensive processes.

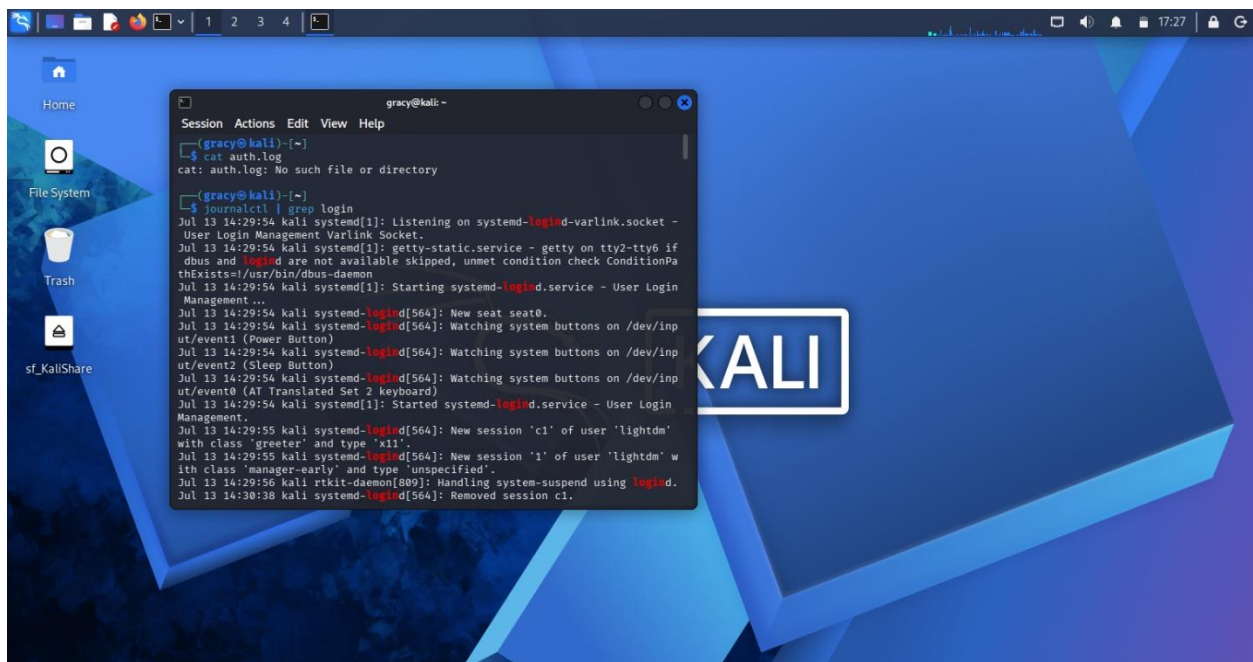
Task 6: Log Analysis

Command 1: cd /var/log

Explanation: Changes the current directory to /var/log, which stores Linux system log files.

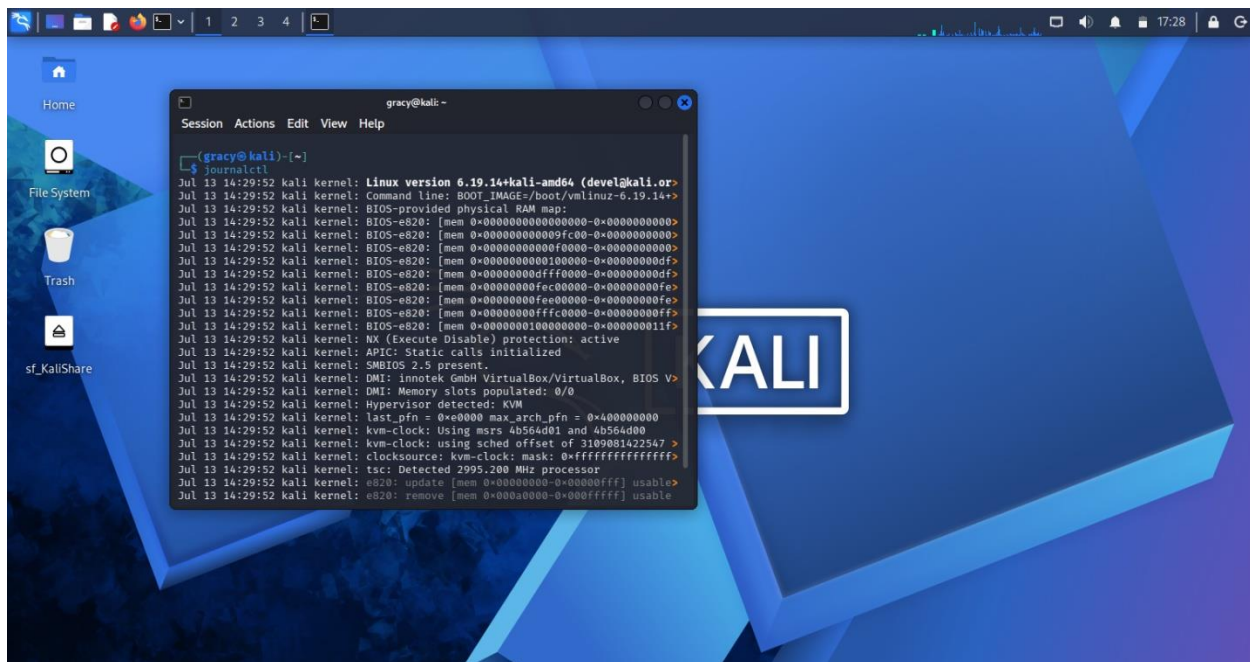


Command 2: cat auth.log



Explanation: Displays authentication logs, including login attempts, user authentication, and sudo activities.

Command 3: journalctl



Explanation: Displays system logs maintained by the systemd journal, including boot messages, login events, and system service logs.

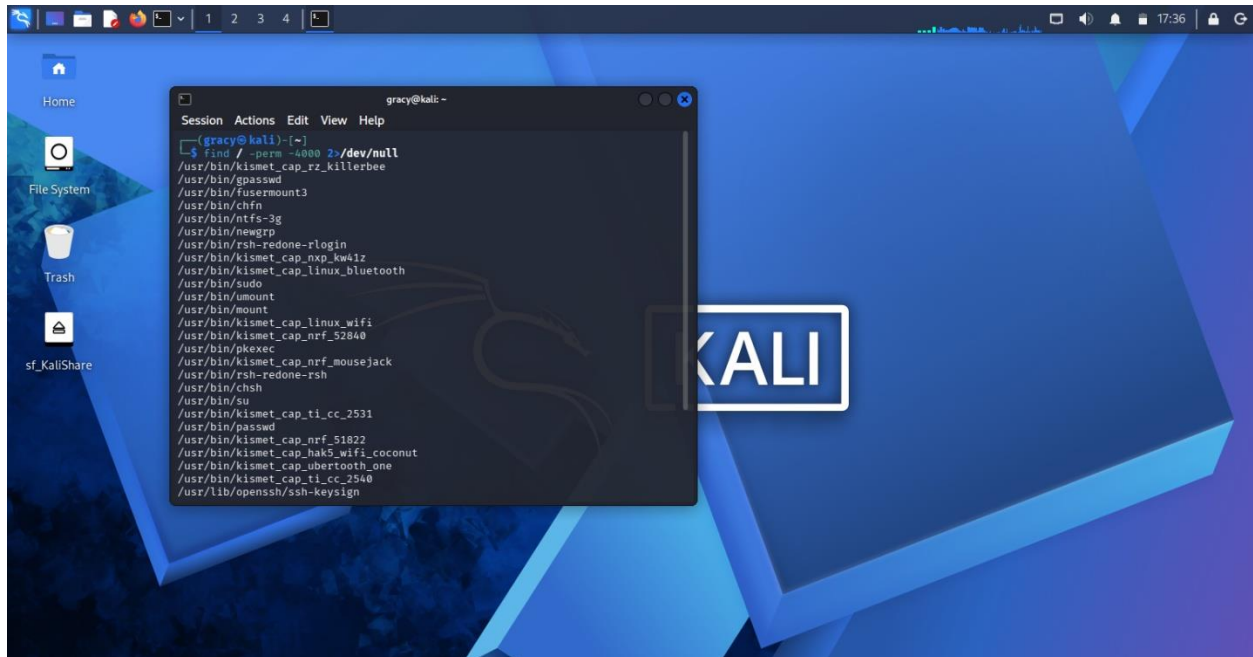
Identification

- **Login Events:** The logs contain records of user login sessions and login attempts.
- **Authentication Logs:** The logs record user authentication, sudo usage, and other security-related activities.

Observation: The log files were successfully accessed from the /var/log directory. The auth.log and journalctl commands displayed login events and authentication-related information, which are useful for monitoring user activity and troubleshooting security issues.

Task 7: Security Audit

Command: `find / -perm -4000 2>/dev/null`



Explanation:

- Searches the entire Linux system for files with the **SUID (Set User ID)** permission.
- `2>/dev/null` hides permission denied and other error messages, making the output easier to read.

What are SUID Files?

SUID (Set User ID) files are executable files that run with the permissions of the file owner instead of the user who executes them. If the file owner is **root**, the program runs with root privileges.

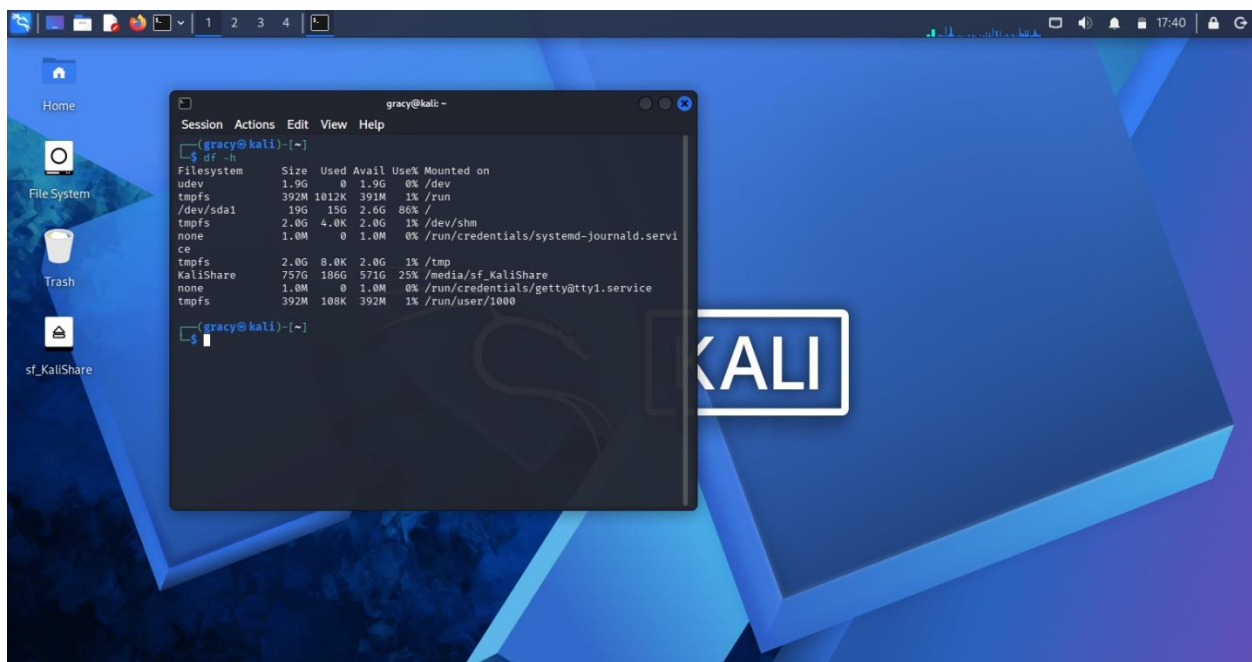
Why are SUID Files a Security Risk?

- They can provide elevated (higher) privileges during execution.
- Vulnerable or misconfigured SUID files may be exploited by attackers to gain unauthorized access.
- Regular security audits help identify and review SUID files to reduce security risks.

Observation: The command successfully listed all files with the SUID permission on the system. These files should be reviewed regularly because they execute with elevated privileges and may become a security risk if they are vulnerable or misconfigured.

Task 8: Basic Troubleshooting

Command 1: df -h



The screenshot shows a Kali Linux desktop with a blue and purple geometric background. A terminal window titled 'gracy@kali ~' is open, displaying the output of the 'df -h' command. The output is a table showing disk space usage for various filesystems. The desktop also features icons for 'Home', 'File System', 'Trash', and 'sf_KaliShare'.

```

gracy@kali ~
$ df -h

```

Filesystem	Size	Used	Avail	Use%	Mounted on
udev	1.9G	0	1.9G	0%	/dev
tmpfs	392M	1012K	391M	1%	/run
/dev/sda1	19G	15G	2.6G	86%	/
tmpfs	2.0G	4.0K	2.0G	1%	/dev/shm
none	1.0M	0	1.0M	0%	/run/credentials/systemd-journald.service
tmpfs	2.0G	8.0K	2.0G	1%	/tmp
KaliShare	757G	186G	571G	25%	/media/sf_KaliShare
none	1.0M	0	1.0M	0%	/run/credentials/getty@tty1.service
tmpfs	392M	108K	392M	1%	/run/user/1000

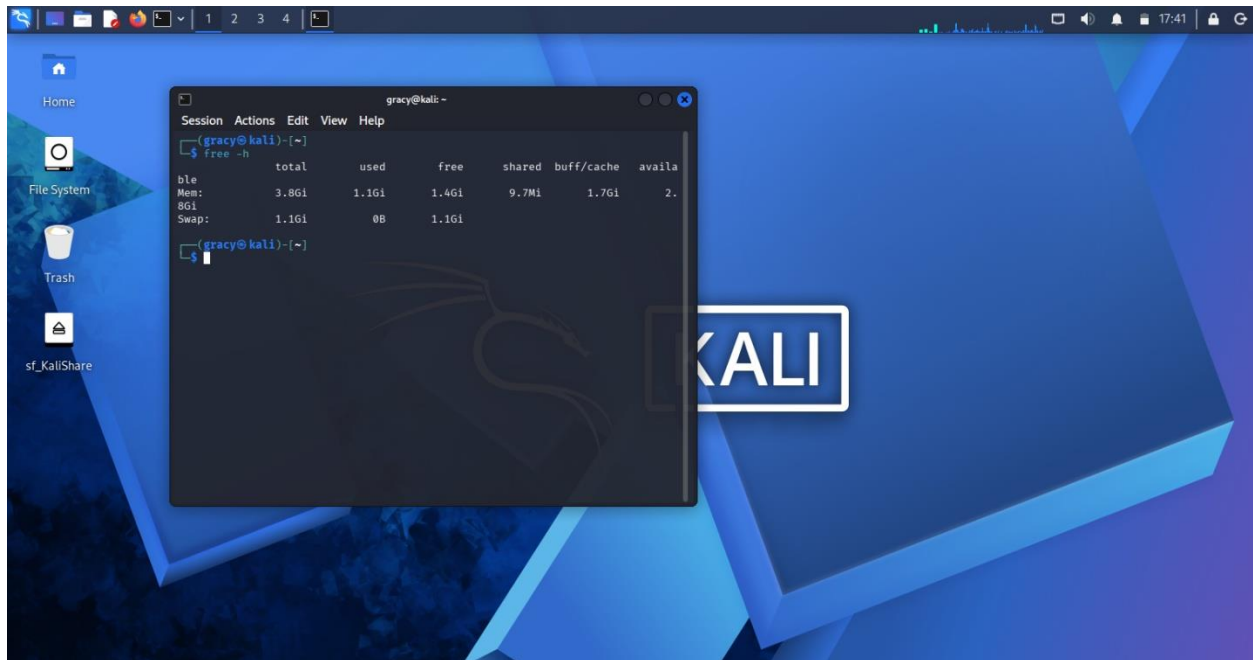
```

gracy@kali ~
$

```

Explanation: Displays disk space usage of all mounted file systems in a human-readable format (KB, MB, GB).

Command 2: `free -h`

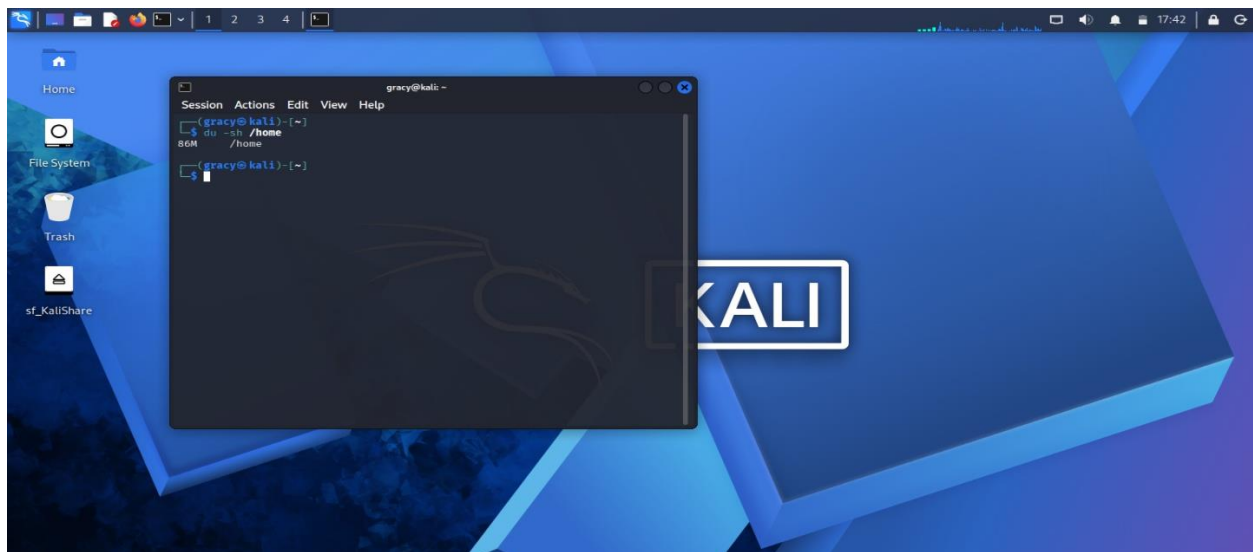


The screenshot shows a Kali Linux desktop with a blue geometric background. A terminal window is open, displaying the output of the `free -h` command. The output is a table showing memory usage for RAM and swap space in human-readable units.

```
gracy@kali: ~  
$ free -h  
              total        used        free      shared  buff/cache   availa  
ble  
Mem:           3.8Gi       1.1Gi       1.4Gi       9.7Mi       1.7Gi       2.  
8Gi  
Swap:          1.1Gi          0B       1.1Gi
```

Explanation: Displays information about system memory (RAM) and swap memory in a human-readable format.

Command 3: `du -sh /home`



The screenshot shows the same Kali Linux desktop environment. The terminal window now displays the output of the `du -sh /home` command, which shows the disk usage of the /home directory in a human-readable format.

```
gracy@kali: ~  
$ du -sh /home  
86M /home  
$
```

Explanation: Displays the total size of the /home directory in a human-readable format.

Observation:

- The df -h command displayed the available and used disk space of the file system.
- The free -h command showed the total, used, and available system memory and swap memory.
- The du -sh /home command displayed the total disk space used by the /home directory.
- These commands help identify disk usage, memory utilization, and storage consumption, making them useful for basic system troubleshooting.

Overall Project Observation:

- All Linux commands were executed successfully, and the expected outputs were obtained.
- The project helped in understanding system information, file system navigation, users and groups, file permissions, process monitoring, log analysis, security auditing, and basic troubleshooting.
- System logs and running processes were successfully analyzed, and the system resources such as disk space and memory usage were monitored.

- The security audit identified SUID files, highlighting the importance of reviewing privileged files to maintain system security.

Challenges Faced:

- Some commands required administrative (sudo) privileges to access system files.
- Understanding Linux file permissions and SUID concepts required careful analysis.
- Large command outputs, such as `top`, `journalctl`, and `cat /etc/passwd`, were difficult to capture in a single screenshot.
- Interpreting system logs and identifying relevant security information required attention to detail.

Conclusion: This project provided practical knowledge of Linux system administration and security auditing. It improved understanding of Linux commands, file permissions, user management, process monitoring, log analysis, and basic troubleshooting techniques. The project also highlighted the importance of regular security audits and system monitoring in maintaining a secure and efficient Linux environment.

